Lab Assignment-3

School of Engineering and Technology
Course Name & Code: Fundamentals of Operating System Lab (ENCA252)
Program Name: BCA (AI & DS) (Research)

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Course – BCA(AI & DS)
Section – A

Problem Title:

Comprehensive Implementation of Page Replacement Algorithms

Problem Statement

In modern operating systems, virtual memory allows efficient utilization of physical memory. When memory frames are full, page replacement algorithms are used to decide which page should be replaced.

This assignment focuses on implementing and analyzing **all major page replacement algorithms**, including **FIFO, LRU, Optimal, MRU, and Second Chance (Clock Algorithm)**. Students will simulate memory behavior and evaluate performance based on page faults.

Learning Objectives

- Understand virtual memory and paging
- Implement multiple page replacement algorithms
- Calculate page faults
- Compare efficiency of algorithms

Tools/Technology Used

- Python 3.x
- Basic Python libraries
- Any IDE (VS Code / PyCharm / IDLE)
- Linux/Ubuntu

Assignment Tasks

Task 1: Input and Page Reference String

Objective:

To understand page request handling.

Description:

Students will input:

- Number of frames
- Page reference string **Sub-Tasks:**

1. Input number of frames
2. Input page sequence
3. Display reference string

Expected Outcome:

- Understanding of paging system
- Ability to simulate memory references

```
(venv) bca_ai_ds_a@ubuntu:~/Downloads$ python ass3.py
Enter number of frames: 4
Enter page reference string: 1 2 3 4 2 1 5 6 2 1 2 3 7 6 3 2 1 2 3 6

Page Reference String:
[1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6]
```

Task 2: FIFO (First-In-First-Out)

Objective:

To implement FIFO algorithm.

Description:

Oldest page is replaced first.

Sub-Tasks:

1. Maintain queue of pages
2. Replace oldest page
3. Count page faults **Expected Outcome:**

- Understanding simple replacement
- Page fault calculation

```
--- FIFO ---
Page 1 -> [1]
Page 2 -> [1, 2]
Page 3 -> [1, 2, 3]
Page 4 -> [1, 2, 3, 4]
Page 2 -> [1, 2, 3, 4]
Page 1 -> [1, 2, 3, 4]
Page 5 -> [2, 3, 4, 5]
Page 6 -> [3, 4, 5, 6]
Page 2 -> [4, 5, 6, 2]
Page 1 -> [5, 6, 2, 1]
Page 2 -> [5, 6, 2, 1]
Page 3 -> [6, 2, 1, 3]
Page 7 -> [2, 1, 3, 7]
Page 6 -> [1, 3, 7, 6]
Page 3 -> [1, 3, 7, 6]
Page 2 -> [3, 7, 6, 2]
Page 1 -> [7, 6, 2, 1]
Page 2 -> [7, 6, 2, 1]
Page 3 -> [6, 2, 1, 3]
Page 6 -> [6, 2, 1, 3]
```

Task 3: LRU (Least Recently Used)

Objective:

To implement LRU algorithm.

Description:

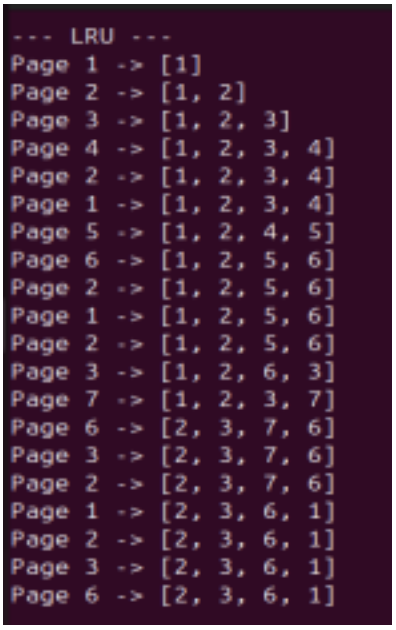
Least recently used page is replaced.

Sub-Tasks:

1. Track usage history
2. Replace least recently used page
3. Count faults

Expected Outcome:

- Understanding temporal locality
- Efficient page tracking

- 

```
--- LRU ---
Page 1 -> [1]
Page 2 -> [1, 2]
Page 3 -> [1, 2, 3]
Page 4 -> [1, 2, 3, 4]
Page 2 -> [1, 2, 3, 4]
Page 1 -> [1, 2, 3, 4]
Page 5 -> [1, 2, 4, 5]
Page 6 -> [1, 2, 5, 6]
Page 2 -> [1, 2, 5, 6]
Page 1 -> [1, 2, 5, 6]
Page 2 -> [1, 2, 5, 6]
Page 3 -> [1, 2, 6, 3]
Page 7 -> [1, 2, 3, 7]
Page 6 -> [2, 3, 7, 6]
Page 3 -> [2, 3, 7, 6]
Page 2 -> [2, 3, 7, 6]
Page 1 -> [2, 3, 6, 1]
Page 2 -> [2, 3, 6, 1]
Page 3 -> [2, 3, 6, 1]
Page 6 -> [2, 3, 6, 1]
```

Task 4: Optimal Page Replacement

Objective:

To implement Optimal algorithm.

Description:

Replaces page that will not be used for longest time in future.

Sub-Tasks:

1. Look ahead in reference string
2. Select page with farthest future use
3. Count page faults

Expected Outcome:

- Minimum page faults
- Benchmark comparison

```
--- OPTIMAL ---
Page 1 -> [1]
Page 2 -> [1, 2]
Page 3 -> [1, 2, 3]
Page 4 -> [1, 2, 3, 4]
Page 2 -> [1, 2, 3, 4]
Page 1 -> [1, 2, 3, 4]
Page 5 -> [1, 2, 3, 5]
Page 6 -> [1, 2, 3, 6]
Page 2 -> [1, 2, 3, 6]
Page 1 -> [1, 2, 3, 6]
Page 2 -> [1, 2, 3, 6]
Page 3 -> [1, 2, 3, 6]
Page 7 -> [2, 3, 6, 7]
Page 6 -> [2, 3, 6, 7]
Page 3 -> [2, 3, 6, 7]
Page 2 -> [2, 3, 6, 7]
Page 1 -> [2, 3, 6, 1]
Page 2 -> [2, 3, 6, 1]
Page 3 -> [2, 3, 6, 1]
Page 6 -> [2, 3, 6, 1]
```

Task 5: MRU (Most Recently Used)

Objective:

To implement MRU algorithm.

Description:

Most recently used page is replaced.

Sub-Tasks:

1. Track recent pages
2. Replace most recently used
3. Count faults

Expected Outcome:

- Understanding reverse locality
- Special-case performance analysis

```
--- MRU ---
Page 1 -> [1]
Page 2 -> [1, 2]
Page 3 -> [1, 2, 3]
Page 4 -> [1, 2, 3, 4]
Page 2 -> [1, 2, 3, 4]
Page 1 -> [1, 2, 3, 4]
Page 5 -> [2, 3, 4, 5]
Page 6 -> [2, 3, 4, 6]
Page 2 -> [2, 3, 4, 6]
Page 1 -> [3, 4, 6, 1]
Page 2 -> [3, 4, 6, 2]
Page 3 -> [3, 4, 6, 2]
Page 7 -> [4, 6, 2, 7]
Page 6 -> [4, 6, 2, 7]
Page 3 -> [4, 2, 7, 3]
Page 2 -> [4, 2, 7, 3]
Page 1 -> [4, 7, 3, 1]
Page 2 -> [4, 7, 3, 2]
Page 3 -> [4, 7, 3, 2]
Page 6 -> [4, 7, 2, 6]
```

Task 6: Performance Comparison

Objective:

To compare all algorithms.

Description:

Students will compare:

- Total page faults
- Efficiency **Sub-Tasks:**

1. Execute all algorithms
2. Compare page faults
3. Identify best algorithm

Expected Outcome:

- Analytical understanding
- Performance comparison

```
--- SECOND CHANCE ---
Page 1 -> [1, -1, -1, -1]
Page 2 -> [1, 2, -1, -1]
Page 3 -> [1, 2, 3, -1]
Page 4 -> [1, 2, 3, 4]
Page 2 -> [1, 2, 3, 4]
Page 1 -> [1, 2, 3, 4]
Page 5 -> [5, 2, 3, 4]
Page 6 -> [5, 6, 3, 4]
Page 2 -> [5, 6, 2, 4]
Page 1 -> [5, 6, 2, 1]
Page 2 -> [5, 6, 2, 1]
Page 3 -> [3, 6, 2, 1]
Page 7 -> [3, 7, 2, 1]
Page 6 -> [3, 7, 6, 1]
Page 3 -> [3, 7, 6, 1]
Page 2 -> [3, 7, 6, 2]
Page 1 -> [1, 7, 6, 2]
Page 2 -> [1, 7, 6, 2]
Page 3 -> [1, 3, 6, 2]
Page 6 -> [1, 3, 6, 2]
```

Task 7: Result Analysis

Objective:

To interpret results.

Description:

Students will:

- Identify best and worst algorithms
- Justify results **Sub-Tasks:**

1. Compare outputs
2. Analyze pros/cons
3. Write conclusion **Expected Outcome:**

- Concept clarity
- Real-world understanding

```
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PAGE FAULTS COMPARISON
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FIFO: 14
LRU: 10
Optimal: 8
MRU: 13
Second Chance: 14

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ANALYSIS
================================
Best Algorithm: Optimal (Least Page Faults)
Worst Algorithm: FIFO (Most Page Faults)
```